

CTseg: A Tool for Brain CT Segmentation, Spatial Normalisation, and Volumetrics

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Abstract

This paper presents and validates CTseg, a freely available software for brain CT segmentation, spatial normalisation, and volumetrics. CTseg builds on the Multi-Brain generative modelling framework, providing a CT-specific pipeline that produces tissue maps, deformation fields, and brain volume estimates in the same format as SPM’s unified segmentation, thereby extending SPM’s established analysis chain from MRI to CT. CTseg is designed for routine hospital CT scans without requiring preprocessing or resampling in deployment. Although CTseg has been adopted in clinical research spanning, among other things, stroke, dementia, and brain morphometry, a systematic validation against an independent reference standard has been lacking. Using paired MR/CT head scans, we evaluate CTseg across four dimensions: segmentation accuracy against an MRI-derived silver standard; spatial normalisation consistency through group-average sharpness and voxelwise coefficient of variation; brain volume agreement via intraclass correlation and Bland-Altman analysis; and downstream sex classification performance from normalised tissue maps. As a baseline, we apply SPM’s MRI-based unified segmentation directly to the CT images. CTseg significantly outperformed this baseline for segmentation and normalisation, showed stronger TBV agreement, and achieved comparable TIV agreement. CTseg is freely available at <https://github.com/WCHN/CTseg>, and all experiment code is included in the repository for full reproducibility.

1 Introduction

Computed tomography (CT) is the most frequently acquired neuroimaging modality in clinical practice. It is the first-line investigation for stroke, traumatic brain injury, and intracranial haemorrhage, and is routinely obtained in emergency, neurosurgical, and oncological settings. Across most healthcare systems, head CT exams substantially outnumber brain MRI exams [Smith-Bindman et al., 2019], yet the dominant computational neuroimaging analysis tools have been developed for MRI.

Being able to automatically segment brain tissue and spatially normalise CT scans to a standard coordinate system [Friston et al., 1995] would unlock significant clinical and research value. It would enable population-level analyses such as voxel-based morphometry [Ashburner and Friston, 2000] on the large volumes of CT data that already exist in hospital archives, support quantitative assessment of brain atrophy from routine clinical scans, facilitate lesion mapping in standard space for stroke and trauma studies, and provide automated brain volume estimates (total brain volume, TBV, and total intracranial volume, TIV) that are recognised as clinical biomarkers. These capabilities are well established for MRI, where software suites such as SPM [Tierney et al., 2025], FreeSurfer [Fischl, 2012], and FSL [Jenkinson et al., 2012] provide mature, validated pipelines. However, applying these MRI-trained tools to CT data is problematic because the tissue contrast in CT is fundamentally different: the Hounsfield unit scale reflects X-ray attenuation, yielding substantially poorer differentiation between grey matter (GM), white matter (WM), and cerebrospinal fluid (CSF) than MRI.

In this paper, we present and validate CTseg¹, a CT-specific tool that addresses these limitations. CTseg can be used within an SPM-compatible framework, without supervised labels or GPU dependence. Although CTseg has already been adopted in clinical research across diverse populations, a systematic validation against an independent reference standard has not yet been performed. We validate CTseg using 59 paired MR/CT head scans from the SynthRAD2025 challenge [Thummerer et al., 2025]. The paired design enables us to use SPM’s MRI segmentation [Ashburner and Friston, 2005] as a silver standard. As a baseline, we also apply SPM’s unified segmentation directly to the CT images, providing a controlled head-to-head comparison that isolates the

¹Not to be confused with the unrelated “CTSeg” tool by [Adduru et al., 2020], which uses a different approach.

benefit of CT-specific modelling. SPM is the appropriate primary comparator because both CTseg and SPM are unsupervised, model-based methods within the same framework. Deep learning methods for CT segmentation typically require labels and generally address a narrower task: they produce segmentations but not the deformation fields or volumetric outputs needed for spatial normalisation, voxel-based morphometry, or brain volume estimation. A comparison with such methods is therefore outside the scope of this validation but would be valuable future work.

2 Related Work

Automated analysis of brain CT has a shorter history than its MRI counterpart, largely because the poor soft-tissue contrast in CT was long considered insufficient for tissue-level segmentation. Early work focused primarily on spatial normalisation. [Rorden et al. \[2012\]](#) developed stroke-aged CT and MRI atlases and packaged them in the SPM Clinical Toolbox, enabling affine and low-dimensional nonlinear normalisation of clinical CT scans for lesion mapping. [Muschelli \[2020\]](#) later produced a higher-resolution, unbiased CT atlas using an iterative registration procedure, together with coarse ventricle and whole-brain labels. Both approaches are atlas and normalisation resources rather than subject-level CT tissue-segmentation methods.

Tissue segmentation from CT has been addressed by a smaller body of work. [Gupta et al. \[2010\]](#) proposed an early approach for automatic GM, WM, and CSF segmentation from unenhanced CT using adaptive thresholding and anatomical knowledge, and [Kemmling et al. \[2012\]](#) introduced a probabilistic Hounsfield-unit decomposition method using MRI-derived tissue maps and a custom CT atlas. [Manniesing et al. \[2017\]](#) demonstrated that GM and WM can be segmented from 4D contrast-enhanced CT by exploiting temporal averaging across a perfusion series, followed by voxel feature extraction, SVM classification, and atlas-guided geodesic active contours. While effective for contrast-enhanced acquisitions, this method is not applicable to the far more common single-phase non-contrast CT.

More recently, deep learning methods have been applied to the problem. [Srikrishna et al. \[2021\]](#) trained a U-Net on paired CT/MRI data from the Gothenburg H70 Birth Cohort, using MRI-derived FreeSurfer labels as supervision, and achieved Dice scores of 0.79 (GM), 0.82 (WM), and 0.75 (CSF) on held-out CT images. [Cai et al. \[2020\]](#) used a deep-learning model to segment 11 intracranial structures from non-contrast head CT, targeting radiological reporting rather than tissue-class analysis. [Gerken et al. \[2023\]](#) trained a 2D U-Net for brain parenchyma and ventricular segmentation on CT, demonstrating robustness to pathology including haemorrhage and tumour. [Son et al. \[2024\]](#) used perceptual-loss U-Nets trained on 199 paired CT/MRI scans for GM, WM, and CSF segmentation. Most recently, [Huisman et al. \[2024\]](#) validated SynthSeg, a domain-randomised segmentation tool originally developed for MRI, on CT using 260 paired CT/MRI radiotherapy scans from five centres. These supervised and domain-randomised methods are segmentation-centric and do not themselves provide nonlinear deformation fields or an SPM-compatible end-to-end morphometry pipeline. Separately, [Fielden et al. \[2022\]](#) compared global CT-versus-MRI brain volumetrics using SPM and FSL in 69 paired scans, providing an independent reference point for CT-based volume estimation.

CTseg occupies a distinct position in this landscape: it is a generative model-based method that jointly estimates tissue segmentation and nonlinear spatial normalisation from a single non-contrast CT scan, without requiring labelled training data, contrast enhancement, or preprocessing. Its outputs are compatible with the standard SPM analysis pipeline, bridging the gap between CT acquisition and the quantitative analysis tools that are well established for MRI.

2.1 Prior Applications of CTseg

Although CTseg was initially released without a formal validation study, it has since been adopted in a range of clinical research settings. Table 1 summarises 13 published studies, collectively encompassing over 5,000 subjects, that have used CTseg or CTseg-derived outputs. Applications span stroke outcome prediction and perfusion mapping, brain morphometry in Down syndrome, COVID-19 neuroimaging, cardiac arrest prognosis, radiomics, PET imaging, bone density estimation, mortality prediction, and dementia screening. These studies demonstrate the practical utility of CTseg across diverse clinical populations, but they do not substitute for direct validation against an independent reference standard – the present study provides such a validation.

Table 1: Published external applications of CTseg, as of April 2026.

Study	Domain	CTseg Use	<i>N</i>	Key Finding
Liu et al. [2021]	Basilar artery occlusion	BPV, CSF volume, brain atrophy index	231	Severe brain atrophy (estimated from CT via CTseg) independently predicted poor functional outcome after endovascular treatment in the BASILAR registry.
Hoving et al. [2024]	Ischaemic stroke	ICV, TBV from baseline NCCT	200	Adjusting CT perfusion core volume by CTseg-derived ICV or TBV did not improve prognostic value beyond absolute core volume in the MR CLEAN Registry.
Luijten et al. [2026]	Ischaemic stroke (EVT)	Brain parenchymal fraction	1,391	CT-based prediction model incorporating CTseg-derived brain atrophy improved identification of patients benefiting from thrombectomy across 7 randomised trials.
Sánchez-Moreno et al. [2024]	Down syndrome	VBM, tissue volumes, spatial normalisation	98	First CT-based VBM study in Down syndrome; CTseg-derived GM volumes correlated with dementia severity and plasma AD biomarkers (p-tau181, NfL) in hippocampal and cingulate regions.
Sánchez-Moreno et al. [2025]	Down syndrome	CT vs MRI morphometry comparison	23	92.3% agreement between CTseg (CT) and SPM DARTEL (MRI) morphometry; similar patterns of regional volume loss associated with cognitive decline.
Urbanos et al. [2025]	Subarachnoid haemorrhage	GM/WM segmentation for radiomics feature extraction	403	Radiomics features extracted from CTseg tissue segmentations predicted 6-month mortality, functional outcome (GOS), vasospasm, and hydrocephalus in SAH patients.
Kalc et al. [2024]	Bone mineral density	CT skull segmentation for validation	114	CTseg skull segmentation of OASIS-3 CT scans used as ground truth to validate MRI-derived proxy measures of bone mineral density and subcutaneous adiposity.
Tsui et al. [2023]	Acute hospital mortality	TBV, TIV from routine NCCT	804	CT-derived brain volumes integrated into a multidimensional clinical model predicted mortality in acutely hospitalised older patients, demonstrating prognostic value of routinely acquired CT data.
Gramespacher et al. [2024]	Cardiac arrest outcome	GM segmentation from NCCT	132	CTseg GM segmentation of cerebral CT used as input to a supervised ML classifier predicting neurological outcome after out-of-hospital cardiac arrest.
Tangwiriyasakul et al. [2026]	Ischaemic stroke perfusion	Spatial normalisation, GM/WM atlases	1,393	CTseg used to co-register and nonlinearly normalise routine NCCT and CTA to MNI space; CTseg GM/WM atlases used to segment derived perfusion-deficit substrates in a deep generative model.

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Study	Domain	CTseg Use	N	Key Finding
Duan et al. [2021]	COVID-19 neuroimaging	VBM, tissue volumes, spatial normalisation	120	CT images normalised to MNI and segmented into six tissue classes using CTseg; VBM revealed frontal-temporal GM volume reductions associated with clinical severity in older COVID-19 patients.
Tonietto et al. [2024]	Alzheimer’s tau PET imaging	Skull/meninges segmentation from CT	18	CTseg skull probability maps from PET attenuation-correction CT used to define skull/meninges ROI for isolating off-target tau tracer binding in a head-to-head comparison of [^{18}F]PI-2620 and [^{18}F]RO948.
Yelanchezian et al. [2025]	Dementia screening	TBV, hippocampal volume	168	CTseg-derived TBV and hippocampal volume from routine clinical CT differentiated dementia from non-dementia (combined AUC = 0.74) in a New Zealand memory service.
Total			5,095	

3 Methods

3.1 Overview

CTseg is a CT-specific pipeline for simultaneous tissue segmentation and nonlinear spatial normalisation. It builds on the Multi-Brain generative modelling framework [Brudfors et al., 2020]², which extends SPM’s unified segmentation approach [Ashburner and Friston, 2005] with improved affine and nonlinear registration and jointly learned atlas and intensity priors. The CTseg model produces six tissue class segmentations (GM, WM, CSF, Bone, Soft tissue, and Background) together with nonlinear deformation fields that map each scan to an atlas space. In addition, CTseg provides skull-stripping and estimates of TBV and TIV.

CTseg works directly on clinical CT scans without preprocessing. It handles arbitrary voxel sizes (including the variable slice thickness common in clinical CT), arbitrary fields of view, and does not assume any particular orientation or prior alignment. Because CTseg produces outputs in the same format as SPM’s unified segmentation (native and atlas-space tissue maps, deformation fields, and modulated images), it serves as a drop-in replacement that extends SPM’s downstream analysis tools, including voxel-based morphometry [Ashburner and Friston, 2000], atlas-based region-of-interest analysis [Tzourio-Mazoyer et al., 2002], and voxel-based lesion-symptom mapping [Bates et al., 2003], from MRI to CT data.

3.2 Training Data

The CTseg atlas and intensity priors were learned jointly from both CT and MRI data. The training set comprised four publicly available datasets:

- **CQ500** [Chilamkurthy et al., 2018]: 491 non-contrast head CT scans, of which the 222 scans reported as normal by all three radiologist readers (no haemorrhage, fracture, mass effect, or midline shift) were used for atlas construction. In-plane resolution ~ 0.5 mm, variable slice thickness (0.625–5.0 mm).
- **IXI**³: 577 multi-sequence MR images (T1w, T2w, PDw) from healthy subjects, acquired at three London hospitals, ~ 1 mm isotropic.
- **MICCAI 2012 Multi-Atlas Labeling** [Landman and Warfield, 2012]: 35 T1-weighted MRI scans (30 unique subjects, 5 rescanned) with expert manual segmentations of 133 brain regions, 1 mm isotropic. The manual labels were combined into GM and WM tissue classes and used to supervise the atlas learning.

²Source code: <https://github.com/WTCN-computational-anatomy-group/mb>

³<https://brain-development.org/ixi-dataset>

- **MRBrainS18** [Kuijff et al., 2024]: 7 training subjects with multi-sequence MRI (T1w, T1-IR, T2-FLAIR) and manual segmentations of 8 brain structures, from subjects aged over 50, including patients with pathologies. The manual labels were combined into GM (cortical grey matter + basal ganglia), WM (white matter + white matter lesions), and CSF (cerebrospinal fluid + ventricles) to supervise the atlas learning. Pathological tissue (e.g. infarctions) was masked out during training.

The inclusion of both modalities is a key design choice: MRI provides superior soft-tissue contrast for learning tissue boundaries, while CT contributes bone structures and ensures the intensity priors capture Hounsfield-unit profiles. Where manual labels were available (MICCAI 2012, MRBrainS18), they guided the merging of learned Gaussian components into tissue classes; the remaining datasets (CQ500, IXI) contributed unlabelled images, with tissue classes inferred entirely from the generative model. All training images were rigidly aligned to MNI space prior to model fitting, which speeds up convergence of learning the Multi-Brain model.

3.3 Atlas Learning

The Multi-Brain framework jointly estimates a tissue atlas, Gaussian mixture model (GMM) intensity priors, and nonlinear deformation fields within a single generative model. The model was configured with 1mm atlas voxel size and $K = 18$ Gaussian components, and fit to the pooled training data using default hyper-parameters for the diffeomorphic registration regularisation, bias field correction, etc. After convergence, the 18 components were merged by visual inspection into six tissue classes. Each class is represented by a single spatial prior (one channel of the atlas) but may be modelled by multiple Gaussian components in the intensity domain. For example, the background class is captured by a single spatial prior map but six Gaussians, reflecting the distinct intensity profiles of air, surrounding tissue, and scanner-specific artefacts (Figure 1). This GMM-per-class structure allows each tissue class to capture multimodal intensity distributions without requiring additional spatial prior channels. Full details of the model specification, optimisation procedure, and convergence criteria are provided in [Brudfors, 2020]. Learning the CTseg model took 24 hours on a 24-core workstation with 128 GB of RAM.

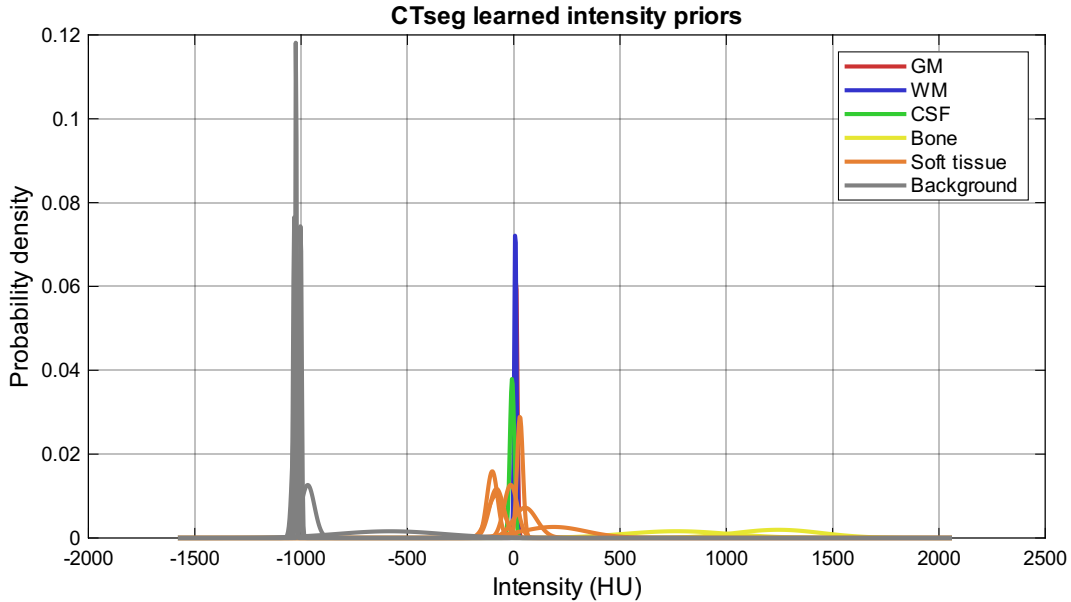


Figure 1: Learned $K = 18$ GMM intensity priors for CT data. Each colour represents one of the six tissue classes. Individual Gaussian components within each class are shown as separate curves of the same colour. For example, the background class (grey) comprises six Gaussian components, reflecting distinct intensity sub-populations (e.g. air, scanner bed).

3.3.1 Original Groupwise Optimal Atlas

The resulting atlas `mu_CTseg.nii` was learned in a data-driven coordinate system that is optimal for the training population (Figure 2). A notable feature is its extended caudal field of view, which includes the upper cervical spine. This benefits processing of clinical CT scans that frequently extend below the skull base into the neck, as

the registration model can anchor to inferior anatomy and reduce misregistration risk when the brain occupies only a portion of the image volume.

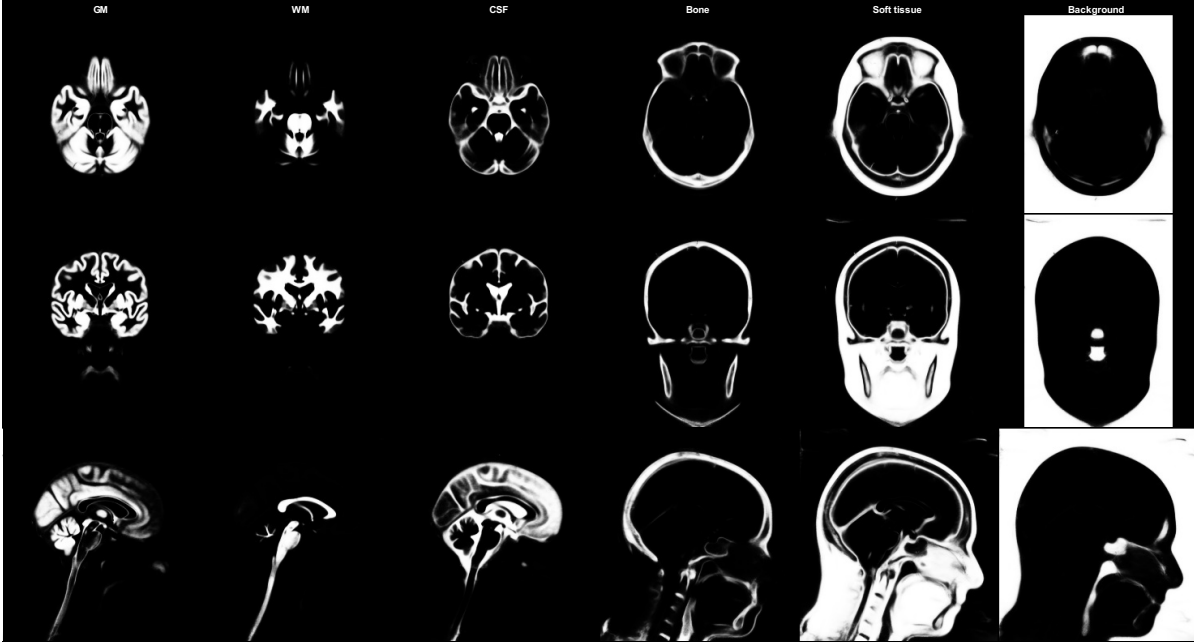


Figure 2: Original groupwise CTseg tissue atlas (`mu_CTseg.nii`), learned in a data-driven coordinate system. Note the extended caudal field of view covering the upper cervical spine. Columns: tissue classes (GM, WM, CSF, Bone, Soft tissue, Background). Rows: axial, coronal, and sagittal views. Brighter voxels indicate higher prior probability.

3.3.2 SPM-aligned Atlases

The CTseg atlas described above was learned in a groupwise-optimal space that does not coincide with the space of the SPM atlas. This is a consequence of the Multi-Brain framework, which learns the atlas jointly with the registration parameters and is therefore not constrained to any predefined coordinate system. While this groupwise space is optimal for the training population, it means CTseg’s deformation fields do not map directly to, e.g., SPM atlas space. To address this, we nonlinearly align the CTseg atlas to SPM’s tissue probability space (`TPM.nii`). Rather than routing the alignment through an intermediate MRI reference image, we register `mu_CTseg.nii` directly to `TPM.nii` with Multi-Brain, using the tissue-probability channels of both atlases as categorical inputs. Because the source and target are already tissue-probability fields, no intensity model is needed: the registration matches posteriors to posteriors, which is both simpler and more faithful than aligning the atlas to a single-subject MR intensity average. The resulting deformation is then applied to warp the CTseg template into SPM space at the requested voxel size.

Two SPM-aligned atlases are distributed with CTseg:

- `mu_CTseg_spm15.nii`: 1.5 mm isotropic resolution, matching the voxel size of `TPM.nii`. This is the default atlas and the version used throughout this paper.
- `mu_CTseg_spm10.nii`: 1.0 mm isotropic resolution, for applications requiring higher spatial fidelity at the cost of longer processing time.

Because the SPM-aligned atlases live directly in SPM space, `wc*/mwc*` outputs can be analysed with standard SPM tooling without an additional warp through `spm_CTseg_warp`.

4 Validation Study

4.1 Data

Validating a CT segmentation tool against an MRI-derived reference requires a dataset of spatially aligned, same-subject MR and CT brain scans. Such datasets are rare in the public domain, as MR and CT are typically acquired

Example paired MR and CT — 1HNA042

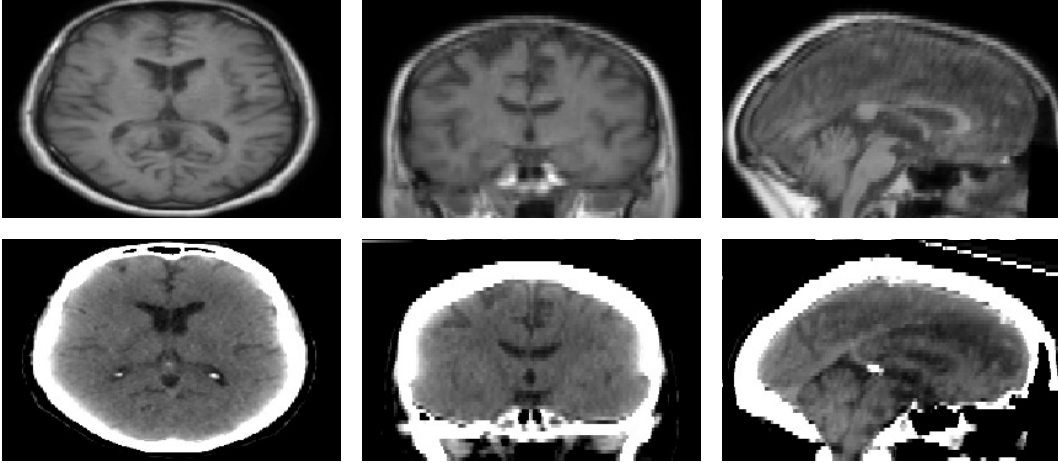


Figure 3: Example paired MR (top) and CT (bottom) images from the SynthRAD2025 dataset (Task 1 Head & Neck, MRI-CT training subset), shown in axial, coronal, and sagittal views.

for different clinical indications and CT involves ionising radiation that is difficult to justify for research alone. The few existing paired datasets come from radiotherapy planning, where both modalities are acquired as part of routine care. We therefore used the SynthRAD2025 Grand Challenge dataset [Thummerer et al., 2025], which is, to our knowledge, among the largest publicly available paired MR-CT datasets. The SynthRAD2025 dataset comprises 2,362 paired MRI-CT and CBCT-CT cases from head-and-neck, thoracic, and abdominal cancer patients, acquired at five European university medical centres (UMC Groningen, UMC Utrecht, Radboud UMC, LMU University Hospital Munich, and University Hospital of Cologne) using diverse scanners and acquisition protocols. From this dataset, we used the Task 1 (Head & Neck) MRI-CT training subset. The full training set comprises 221 subjects from three centres (A, C, and D); however, centre D data (65 subjects) is distributed under a restricted licence that limits its use to the duration of the challenge, so we used the 156 freely available subjects from centres A ($n=91$) and C ($n=65$), each with a paired MR and CT scan.

The challenge organisers pre-processed all images prior to release [Thummerer et al., 2025]: (1) MR images were rigidly registered to the corresponding planning CT using Elastix; (2) facial structures were removed (defaced) for anonymisation; (3) images were resampled to $1 \times 1 \times 3$ mm voxel spacing; and (4) images were cropped to the patient outline. No bias field correction or intensity normalisation was applied to the MR images. Images were provided in MetaImage format (.mha) and stored as INT16. Notably, deformable registration of the CT to the MR, to correct for anatomical differences between the two scan sessions, was only performed by the challenge organisers for the validation and test sets, not for the training data. We therefore replicated this step ourselves (described in Section 4.2).

The accompanying metadata provides patient demographics and acquisition parameters. Of the 156 subjects, 64 were male, 27 female, and 65 had sex unlisted; age was available for 90 subjects (mean 64 years, range 29-87). CT scans were acquired on scanners from two manufacturers (Philips, $n=118$; Siemens, $n=38$) across four scanner models, at 120 kVp, with slice thicknesses of 1-3 mm. Figure 3 shows example MR and CT views from a representative subject.

4.2 Preprocessing

All images were preprocessed using the `spm-hospital-preproc` toolbox⁴. Preprocessing consisted of four steps: (1) rigid-body alignment of each image to MNI space; (2) co-registration of the MR and CT images within each subject pair using normalised mutual information, to ensure precise spatial correspondence; (3) cropping of both images to the SPM12 atlas field of view ($181 \times 217 \times 60$ voxels at the original $1 \times 1 \times 3$ mm voxel size); and (4) deformable registration of the CT to the MR to correct for residual anatomical differences between the two scan sessions (e.g. from weight change or organ motion), replicating the procedure used by the challenge organisers for the validation and test sets [Thummerer et al., 2025]. The deformable registration used the Elastix framework [Klein

⁴<https://github.com/WTCN-computational-anatomy-group/spm-hospital-preproc>

et al., 2010] via SimpleITK with a B-spline transform (final grid spacing 15 mm, three resolution levels), optimised with Mattes mutual information, a bending energy penalty, and a rigidity penalty, using the same parameter file⁵ as the challenge preprocessing pipeline. These steps ensure that the MR and CT images share the same voxel grid, so that tissue maps produced by different segmentation methods can be compared voxelwise. The resulting deformed CT was used as input for all subsequent CT segmentation steps. We note that this preprocessing was performed solely to enable the paired comparison in this validation study; CTseg itself does not require any such preprocessing when used in practice.

After preprocessing, of the 156 subjects from centres A and C, 97 were excluded during quality review because they had incomplete brain coverage (the head-and-neck radiotherapy field of view often truncates the vertex or skull base). The remaining 59 subjects were processed by all the three below pipelines, where all three methods succeeded for every subject, so no further exclusions were necessary.

4.3 Segmentation Pipelines

Three segmentation pipelines were applied to each subject:

1. **SPM on MRI (silver standard).** SPM12’s unified segmentation [Ashburner and Friston, 2005] was applied to each preprocessed MR image using default parameters, producing native-space probability maps for six tissue classes. We retained the GM, WM, and CSF maps as the silver-standard reference segmentations.
2. **SPM on CT (baseline).** The same SPM12 pipeline was applied to each preprocessed CT image. Because SPM’s tissue priors were designed for MRI, this represents an out-of-domain application. Bias field regularisation was increased (`biasreg=10`) to suppress bias field estimation, since CT images lack the intensity inhomogeneity characteristic of MRI. The forward deformation field was saved and used to produce a spatially normalised CT image via SPM’s normalise-write module.
3. **CTseg on CT.** CTseg was applied to each preprocessed CT image using the `mu_CTseg_spm15.nii` atlas, so that CTseg’s deformation fields map directly into SPM space. Native-space tissue maps and modulated warped tissue maps were retained, and the estimated deformation field was used to produce a spatially normalised CT image.

All three pipelines produced modulated warped GM, WM, and CSF tissue maps in MNI space at 1.5 mm resolution, used for the predictive validation experiment.

4.4 Evaluation

4.4.1 Segmentation accuracy

For each tissue class (GM, WM, CSF), we compared the CT-based segmentations against the MRI silver standard. Tissue probability maps were binarised at a threshold of 0.5 before computing overlap and surface distance metrics. We report three complementary measures:

- **Dice coefficient:** volumetric overlap, ranging from 0 (none) to 1 (perfect).
- **95th percentile Hausdorff distance (HD95):** near-worst-case boundary error (mm), robust to outliers.
- **Average symmetric surface distance (ASSD):** mean boundary displacement (mm).

4.4.2 Spatial normalisation quality

Normalisation quality was assessed in three ways. First, the group-average of all normalised CT images was computed separately for each deformation (SPM-MR, SPM-CT, and CTseg); a sharper average indicates more consistent normalisation. Second, we computed the voxelwise coefficient of variation ($\text{CoV} = \sigma/|\mu|$) across subjects within a brain mask (mean intensity > 20 HU); a lower mean CoV indicates more consistent normalisation. Third, we directly compared the spatial accuracy of the deformation fields by warping each subject’s native-space MR tissue maps (GM, WM, CSF) to atlas space using both the SPM-CT and CTseg deformations, then computing Dice and ASSD against the SPM-MR warped tissue maps (binarised at 0.5) as reference. This isolates the quality of the deformation from differences in the segmentation model, since all three methods warp the same native-space MR tissue maps.

⁵<https://github.com/SynthRAD2025/preprocessing>

4.4.3 Brain volumetrics

TBV (GM + WM) and TIV (GM + WM + CSF) were computed for all three methods by summing native-space tissue probability maps weighted by voxel volume. Before summation, an intracranial mask was applied to exclude voxels outside the brain. The mask was derived from the CTseg atlas by computing the softmax tissue probabilities and thresholding the sum of GM, WM, and CSF at 0.5. This atlas-space mask was then warped to each subject’s native space using the SPM-MR deformation field. The same mask was applied to all three methods to ensure a fair comparison. This step is necessary because head CT scans typically extend well below the skull base, and without masking, tissue probability assigned to extracranial structures (e.g. cervical spine white matter, facial soft tissue) can inflate volume estimates, particularly for SPM, whose tissue priors were designed for brain MRI and may assign non-negligible probabilities to non-brain tissue when applied to CT. Agreement was assessed using ICC(3,1) (a two-way mixed-effects, single-measures intraclass correlation assessing consistency) [Shrout and Fleiss, 1979], Pearson correlation, and Bland-Altman analysis [Bland and Altman, 1986].

4.4.4 Predictive validation of normalisation

As an implicit measure of normalisation quality, we evaluated how well normalised tissue maps support sex classification, following the paradigm of [Brudfors et al., 2019]. The rationale is that better spatial normalisation should yield more anatomically consistent feature representations across subjects, improving discriminability in downstream analyses. CTseg’s diffeomorphic registration model is more expressive than SPM12’s default parameterisation, and its atlas was learned jointly from MRI and CT data; if these richer deformations produce more descriptive normalised tissue maps, this should be reflected in predictive performance, potentially even outperforming SPM applied to MRI. We focus on sex rather than age because sex is discriminable largely from gross morphological size (total brain and intracranial volume), which is preserved directly in the modulated warped tissue maps, making it well-matched to a small-sample ($n = 58$) evaluation of normalisation quality. Age, by contrast, is a continuous target whose signal lies in subtler distributed atrophy patterns and would require a substantially larger cohort to recover reliably from these features alone.

We used $n = 58$ subjects for the experiment (one of the 59 subjects lacked sex metadata and was excluded), comprising 44 males and 14 females (76% male, age range 35–85 years, mean 63.8 ± 11.1). For each method, modulated warped GM, WM, and CSF maps (i.e. tissue probability maps warped to atlas space and scaled by the Jacobian determinant to preserve total tissue volume) were smoothed (8 mm FWHM), masked to intracranial voxels, and concatenated into a single feature vector per subject. These features were used directly (without dimensionality reduction) in a kernel-based Gaussian process (GP) classification framework implemented in PRoNTo [Schrouff et al., 2013], which computes a linear kernel from the feature vectors. Mean centering was performed within each cross-validation fold to prevent information leakage. We used 10-fold cross-validation with identical folds across methods. Folds were assigned at random and were not stratified by sex; given the cohort imbalance (44 male, 14 female), individual folds may contain few or zero female subjects, contributing variance to the per-fold accuracy estimate.

4.4.5 Runtime

Processing time was recorded for each method and subject (segmentation only, excluding I/O and warping).

4.5 Statistical Analysis

Paired Wilcoxon signed-rank tests (two-sided, Bonferroni-corrected for three tissue classes) compared segmentation metrics between SPM-on-CT and CTseg-on-CT; the same tests and correction procedure were applied to the normalisation metrics (warped-tissue Dice and ASSD). Segmentation and normalisation metrics are reported as median [interquartile range]. Volumetric agreement is reported as ICC with 95% confidence intervals, Pearson r , and Bland-Altman bias with 95% limits of agreement. Significance level: $\alpha = 0.05$.

5 Results

5.1 Segmentation Accuracy

Table 2 and Figure 4 summarise the segmentation metrics across all subjects and tissue classes. CTseg achieved significantly higher Dice scores and lower surface distances than SPM-on-CT for all three tissue classes ($p < 0.001$

Table 2: Segmentation accuracy compared to MRI silver standard. Values are median [IQR]. p -values from Wilcoxon signed-rank test (Bonferroni-corrected). Best value per column in bold.

Metric	Method	GM	WM	CSF
Dice	SPM	0.531 [0.515, 0.561]	0.659 [0.647, 0.672]	0.367 [0.342, 0.405]
	CTseg	0.601 [0.588, 0.622]	0.721 [0.710, 0.731]	0.466 [0.439, 0.516]
	p -value	<0.001	<0.001	<0.001
HD95 (mm)	SPM	4.243 [4.123, 4.690]	7.810 [6.500, 8.602]	6.000 [5.385, 6.403]
	CTseg	3.162 [3.000, 3.317]	4.583 [4.243, 4.690]	5.099 [4.610, 6.383]
	p -value	<0.001	<0.001	0.099
ASSD (mm)	SPM	1.135 [1.072, 1.229]	1.818 [1.646, 1.953]	1.821 [1.704, 1.949]
	CTseg	0.865 [0.807, 0.918]	1.078 [1.029, 1.136]	1.513 [1.430, 1.662]
	p -value	<0.001	<0.001	<0.001

for all Dice and ASSD comparisons; CSF HD95 did not reach significance). Figure 5 shows tissue probability maps for a representative subject, and Figure 6 shows segmentation overlays for the best and worst cases.

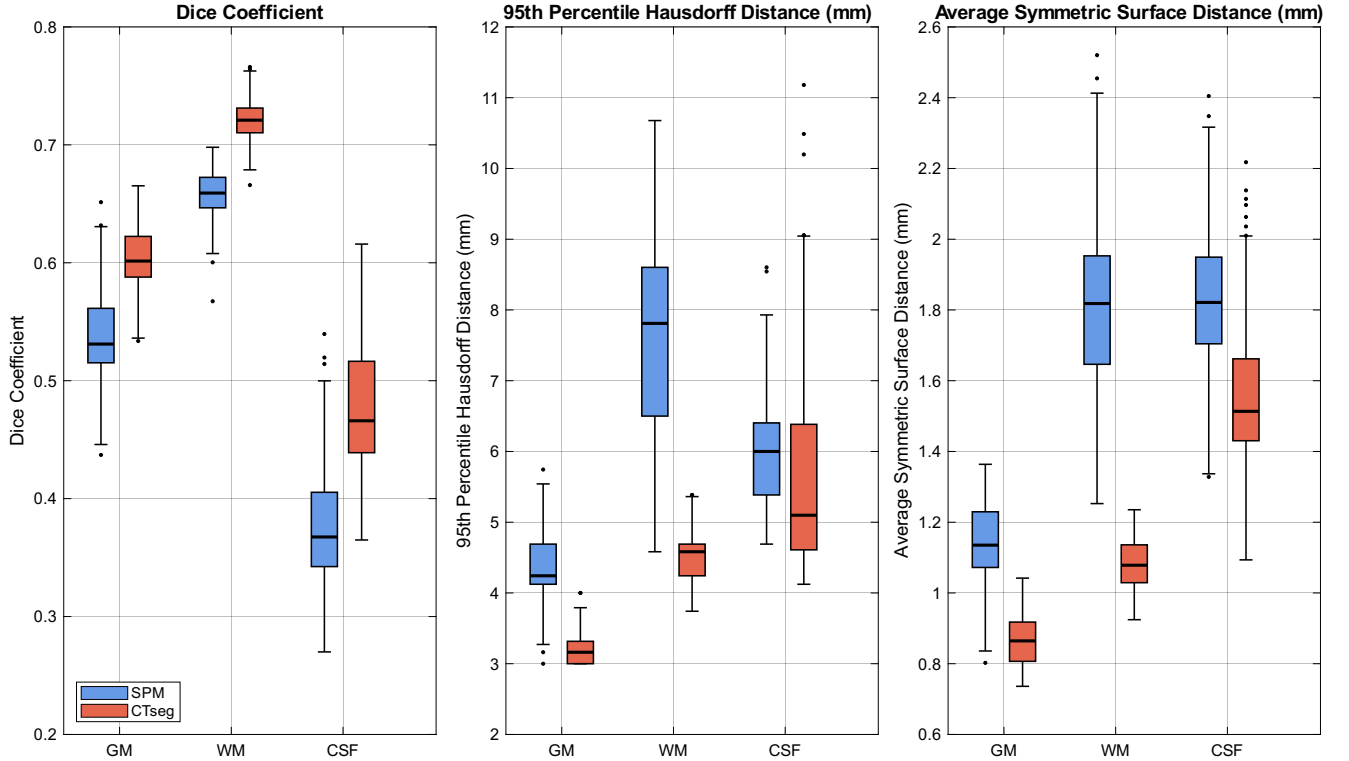


Figure 4: Segmentation metrics comparing SPM and CTseg on CT against the MRI silver standard. Left: Dice coefficient (higher is better). Centre: HD95 (lower is better). Right: ASSD (lower is better).

5.2 Spatial Normalisation

Figure 7 shows the group-average normalised CT images computed using the three deformation fields. Visual inspection suggests that the CTseg average preserves cortical detail and ventricular boundaries more clearly than the SPM-CT average, consistent with more consistent spatial normalisation across subjects. The mean voxelwise coefficient of variation (CoV) within the brain mask was 0.306 for CTseg versus 0.402 for SPM-CT, confirming more consistent alignment. To quantify normalisation accuracy independently of segmentation differences, we warped each subject’s native-space MR tissue maps to atlas space using both the SPM-CT and CTseg deformations, then compared against the SPM-MR warped maps as reference (Figure 8). CTseg achieved significantly higher Dice

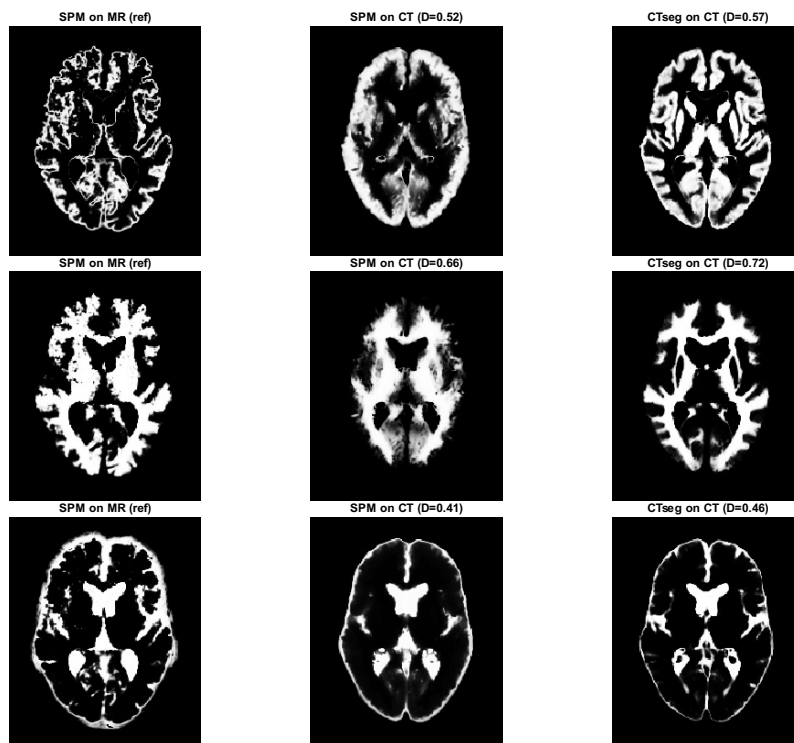


Figure 5: Tissue probability maps for a representative subject (near-median Dice). Rows: GM, WM, CSF. Columns: SPM on MRI (silver standard), SPM on CT, CTseg on CT. Looking at, e.g., GM of CTseg, it clearly shows finer details in the cortex compared to SPM on CT.

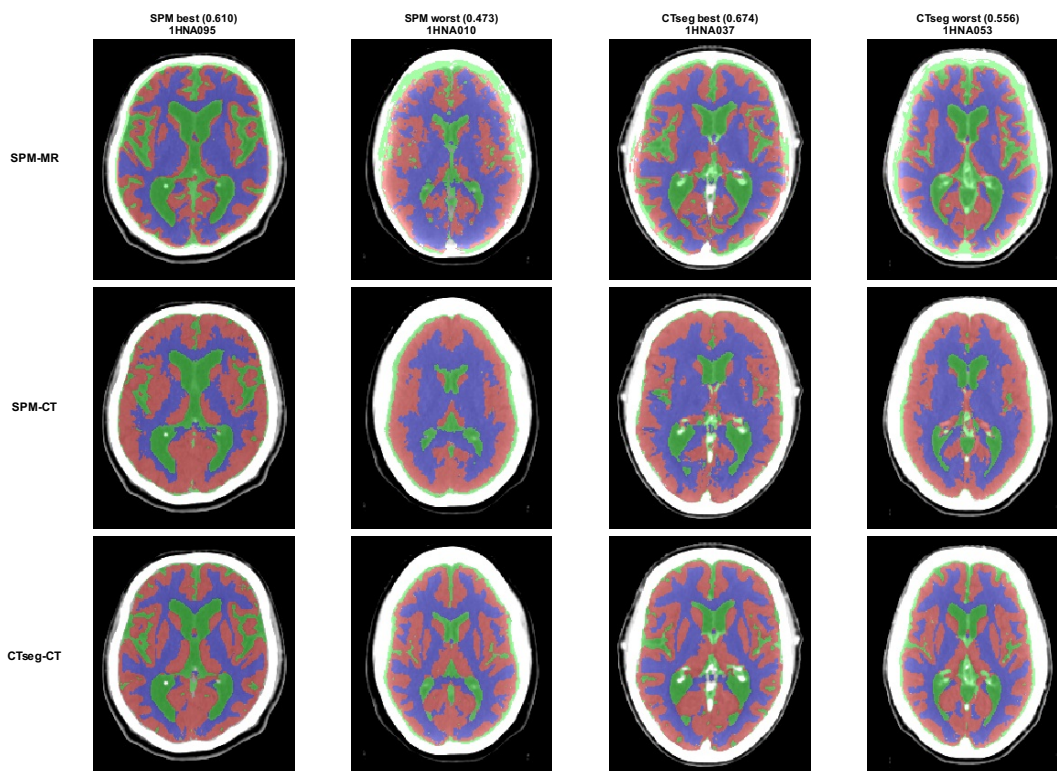


Figure 6: Segmentation overlays for best and worst cases (axial view). Rows: SPM on MRI (silver standard), SPM on CT, CTseg on CT. Columns: best and worst by SPM mean Dice (left pair), best and worst by CTseg mean Dice (right pair). Tissue colours: GM (red), WM (blue), CSF (green). Mean Dice scores are shown in column headers.

overlap than SPM-CT for GM (0.667 vs. 0.633, $p < 0.001$) and WM (0.808 vs. 0.764, $p < 0.001$). For CSF, CTseg and SPM-CT gave essentially identical median Dice (0.636 vs. 0.636, $p = 0.21$). ASSD was significantly lower for CTseg in GM (0.887 vs. 0.970 mm) and WM (0.821 vs. 0.932 mm), with no significant difference for CSF. These results confirm that CTseg’s deformation fields more accurately align brain anatomy to atlas space than those obtained by applying SPM’s MRI-trained model to CT.

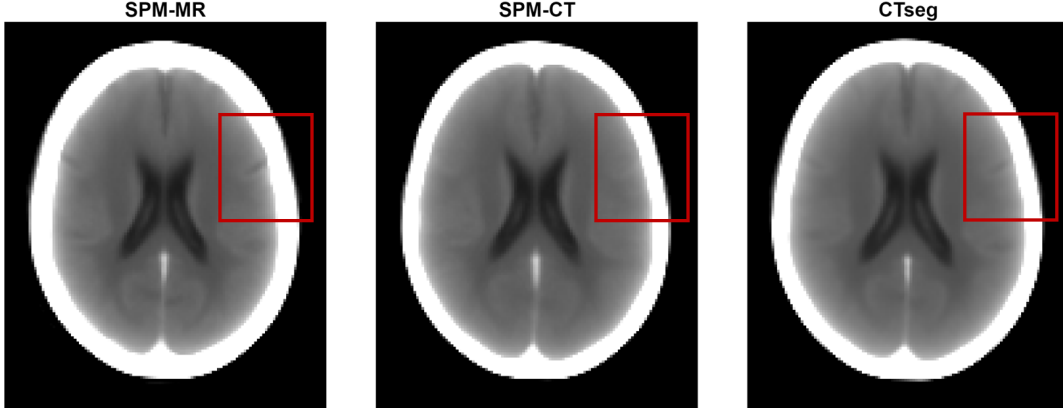


Figure 7: Group-average normalised CT (axial view) for each deformation method. Sharper anatomical boundaries indicate more consistent spatial normalisation across subjects. Red boxes highlight the lateral ventricle and insular cortex region, where the CTseg average preserves cortical sulcal detail and the grey-white matter boundary comparably to the SPM-MR reference, whereas the SPM-CT average is visibly more blurred.

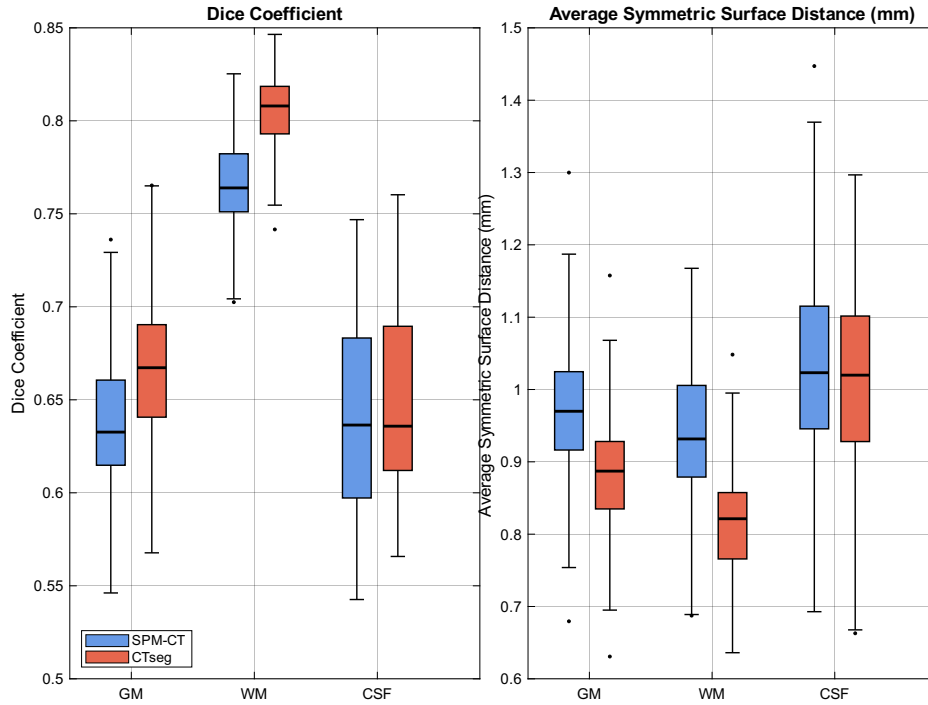


Figure 8: Normalisation metrics comparing SPM-CT and CTseg deformations. Native-space MR tissue maps were warped to atlas space using each deformation and compared against the SPM-MR warped tissue maps as reference. Left: Dice coefficient (higher is better). Right: ASSD (lower is better). HD95 is omitted because at the 1.5 mm atlas resolution, the distance transform quantises surface distances into a few discrete values, making HD95 insensitive to differences between methods.

Table 3: Agreement of brain volume estimates (CT methods vs. MRI reference). Best value per column (within each measure) in bold; for Bias, smaller magnitude is better, for LoA, tighter interval is better.

Measure	Method	ICC [95% CI]	r	Bias (ml)	LoA (ml)
TBV	SPM	0.650 [0.474, 0.776]	0.670	3.8	[-152.8, 160.3]
TBV	CTseg	0.829 [0.729, 0.895]	0.830	-59.1	[-156.1, 37.8]
TIV	SPM	0.798 [0.683, 0.875]	0.807	-196.3	[-330.5, -62.2]
TIV	CTseg	0.821 [0.716, 0.889]	0.823	-229.1	[-349.9, -108.2]

5.3 Brain Volumetrics

Table 3 and Figure 9 report volumetric agreement between CTseg and MRI-derived estimates. For TBV, CTseg achieves a substantially higher ICC (0.829 vs. 0.650) and tighter limits of agreement than SPM-CT, indicating that it better preserves the relative ordering of brain sizes across subjects. CTseg shows a systematic negative bias of approximately -59 ml, whereas SPM-CT’s bias is near zero (3 ml). For TIV, both methods underestimate the MR-derived reference by approximately 196–229 ml, with similar ICC values (SPM-CT: 0.798, CTseg: 0.821). The TIV underestimation is expected given the inherently poor contrast between CSF and brain parenchyma on CT.

If SPM-CT shows closer volumetric agreement with MRI-derived volumes than CTseg does, this should be interpreted with caution: the MRI reference volumes are themselves produced by SPM’s unified segmentation, which shares the same tissue probability atlas, Gaussian mixture structure, regularisation, and cleanup pipeline as SPM-CT. Two runs of the same generative model with the same priors will produce correlated systematic biases, even when the input modality differs. Closer volume agreement between SPM-CT and SPM-MR may therefore reflect shared-method bias rather than genuinely superior accuracy. CTseg, by contrast, uses a different atlas, different priors, and a different registration framework (Multi-Brain), so its volume estimates are more independent of the MRI-derived reference. The observed systematic bias suggests that post hoc calibration may be feasible, but this was not evaluated here; by contrast, the lower correlation of SPM-CT reflects poorer rank-order agreement that cannot be corrected by a linear offset. A further source of volume discrepancy specific to CTseg is its atlas tissue class definitions: structures such as ventricular calcifications (commonly found in the choroid plexus – see CSF classes in Figure 5) and the venous sinuses are modelled as non-brain classes (bone or soft tissue rather than CSF or GM/WM), as visible in the CTseg atlas (see Figure 2). Because these structures are intracranial but assigned to non-brain tissue classes, they are excluded from TBV and TIV estimates, contributing to the systematic negative bias relative to the MRI reference.

5.4 Predictive Validation

For sex classification (Figure 10), CTseg achieved the numerically highest AUC (0.920), with SPM-on-MR (0.886) and SPM-on-CT (0.869) close behind, and the highest balanced accuracy (87.9% vs. 86.2% and 84.5%). AUC differences were not tested for statistical significance given the small, sex-imbalanced cohort (44 male, 14 female; $n = 58$); the results should therefore be read as suggestive rather than confirmatory. The pattern is nevertheless consistent with the hypothesis that CTseg’s more expressive diffeomorphic registration preserves sex-discriminative morphological features (e.g. total brain size, cortical thickness patterns) at least as faithfully as SPM’s parameterisation. AUC is threshold-independent and robust to the imbalance: a classifier that simply predicts the majority class would achieve 76% accuracy but an AUC of only 0.5.

5.5 Runtime

All experiments were run on a workstation with an Intel Core Ultra 7 165H (3.80 GHz) processor and 32 GB RAM, running Windows 11. No GPU acceleration was used. Median segmentation time per subject was 84 s (mean 83 ± 11 s) for SPM-MR, 100 s (mean 103 ± 17 s) for SPM-CT, and 165 s (mean 170 ± 24 s) for CTseg ($n = 59$). CTseg is approximately twice as slow as SPM, which is expected given its more flexible nonlinear registration model (Multi-Brain diffeomorphic framework versus SPM’s DCT-based normalisation). All three methods complete within minutes on a standard workstation without GPU acceleration, making them practical for both research and clinical deployment.

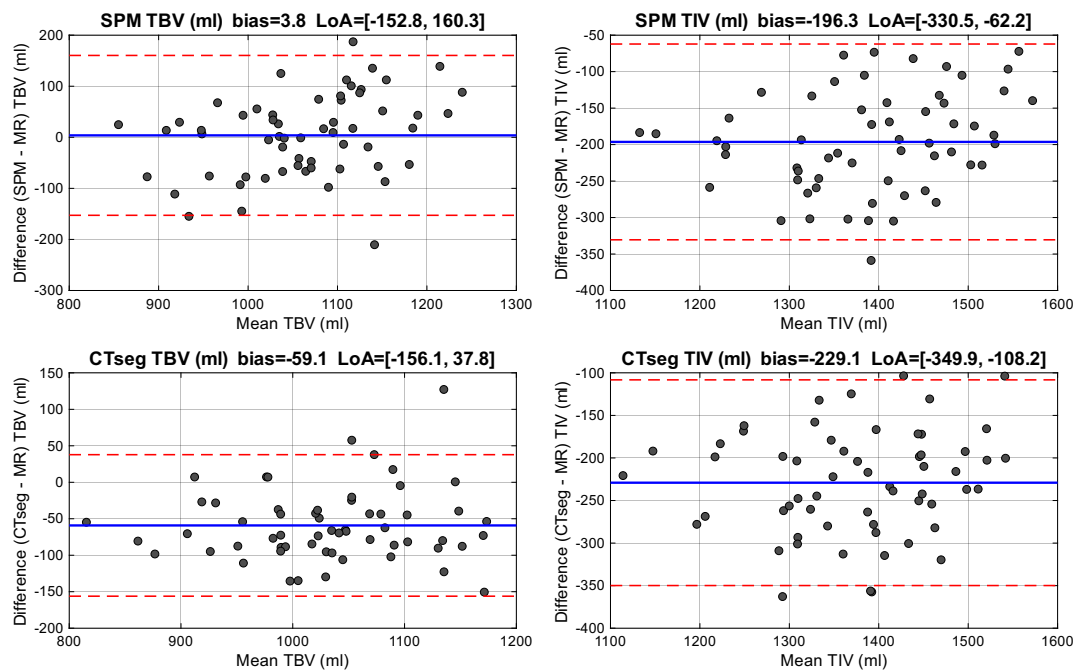


Figure 9: Bland-Altman plots for TBV (left) and TIV (right). Solid line: mean bias. Dashed lines: 95% limits of agreement.

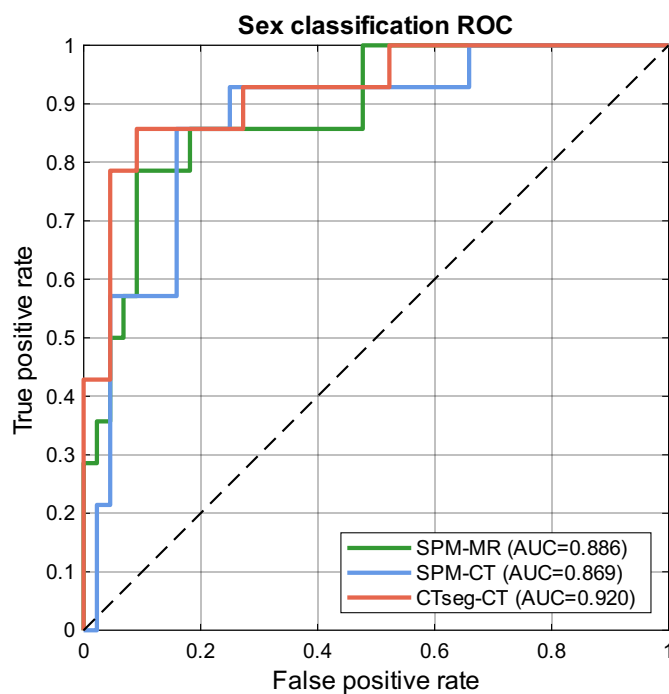


Figure 10: Sex classification ROC curves. Higher AUC indicates better normalisation quality.

6 Discussion

CTseg addresses a different problem from supervised CT labelling tools: it provides a CT-specific generative pipeline for tissue segmentation, spatial normalisation, and brain volumetrics within an SPM-compatible framework, bridging the gap between the large volume of clinical CT data and the quantitative analysis tools established for MRI. The goal of this validation is to demonstrate that CTseg produces reliable outputs across four complementary evaluation dimensions, substantially outperforming the natural baseline of applying SPM’s MRI-trained model to CT. Beyond this formal validation, CTseg’s practical utility is supported by its adoption in over a dozen clinical studies spanning stroke, dementia, brain morphometry, and other domains (Section 2.1).

For segmentation, CTseg achieved significantly higher Dice scores for all three tissue classes (GM: 0.601 vs. 0.531; WM: 0.721 vs. 0.659; CSF: 0.466 vs. 0.367; all $p < 0.001$), with correspondingly lower surface distances. These scores are lower than those reported for supervised deep learning methods trained on paired CT/MRI data [Srikrishna et al., 2021], which is expected: CTseg is an unsupervised generative model that requires no labelled training data and jointly estimates segmentation, registration, and bias field parameters. The trade-off is generality: CTseg is designed to work directly on routine non-contrast CT scans, whereas supervised methods require retraining for new populations, scanners, or acquisition protocols. The largest absolute improvement over SPM-CT was in CSF, where the poor CT contrast between CSF and parenchyma poses the greatest challenge for MRI-trained models. That even GM and WM, which are better differentiated on CT than CSF, show significant improvement underscores the value of CT-specific intensity priors and atlas geometry.

Spatial normalisation quality showed a similar pattern: CTseg produced sharper group averages and higher Dice overlap when the same native-space MR tissue maps were warped using each method’s deformation field. This confirms that the improvement is not merely due to different segmentation models but reflects genuinely more accurate anatomical alignment.

For brain volumetrics, CTseg achieved a substantially higher ICC for TBV (0.829 vs. 0.650), indicating that it better preserves the relative ordering of brain sizes across subjects. CTseg’s systematic negative bias (−59 ml for TBV, −229 ml for TIV) is partially attributable to its atlas class definitions and the inherently poor CSF contrast on CT (discussed further in Limitations). The predictive validation provides a complementary application-level assessment: CTseg achieved the numerically highest sex classification AUC (0.920 vs. 0.869 for SPM-CT and 0.886 for SPM-MR). Given $n = 58$ and the 76%/24% sex imbalance this difference is suggestive rather than statistically established, but the pattern is consistent with CTseg’s more expressive registration model producing normalised tissue maps that preserve sex-discriminative morphological features at least as faithfully as SPM applied to the higher-contrast MRI data.

6.1 Limitations

6.1.1 Study limitations

Our MRI-derived reference segmentations constitute a *silver* standard, not a gold standard; both CT-based methods may be penalised for disagreeing with the MRI reference where it is itself inaccurate. We validate only GM, WM, and CSF, the classes for which MRI provides a credible reference, and skull stripping, although one of CTseg’s outputs, was not directly evaluated. We also do not compare against deep learning-based methods: as discussed in Section 1, no established deep learning tool offers the same joint segmentation, normalisation, and volumetric pipeline, and a comparison between supervised and unsupervised paradigms would be better addressed in dedicated future work.

The SynthRAD2025 dataset consists of head-and-neck radiotherapy planning scans from cancer patients, not a general neuroimaging cohort. Publicly available paired MR-CT brain datasets are exceedingly rare, so co-registered same-subject data almost exclusively comes from radiotherapy workflows; SynthRAD2025 is, to our knowledge, among the largest such resources for the head region. Of the 156 available subjects, 97 (62%) were excluded for incomplete brain coverage (a consequence of the radiotherapy planning field of view, which often truncates the vertex or skull base), leaving 59 for quantitative analysis. This limits statistical power and generalisability, although the successful application of CTseg across diverse clinical populations provides indirect evidence of broader applicability. The predictive validation is limited by the small sample size ($n = 58$) and the imbalanced sex distribution (76% male) in this retained cohort.

6.1.2 Method limitations

CT provides substantially lower contrast between CSF and brain parenchyma than MRI, with the CSF–cortex boundary being particularly ambiguous near cortical sulci and the periventricular margin. CTseg’s strong tissue prior

partially mitigates this, but CSF under-segmentation is an inherent ceiling of CT-based brain segmentation rather than an algorithmic flaw. Since TIV is defined as $GM + WM + CSF$, any CSF under-segmentation directly reduces the TIV estimate, and TBV may also be affected if partial-volume voxels at tissue boundaries are misclassified.

The CTseg atlas was learned from adult brain scans (CQ500, IXI, MICCAI 2012, MRBrainS18). Paediatric and neonatal anatomy, and major pathology not represented in these cohorts (large tumours, extensive haemorrhage, post-surgical cavities, ventriculomegaly well beyond the training distribution), are therefore outside the support of the generative model. Under such distribution shift the unified-segmentation framework can mis-assign voxels to the nearest-intensity tissue class, producing plausible-looking but structurally incorrect segmentations. Users working with populations far from the training distribution should validate on representative cases before trusting CTseg outputs quantitatively.

CTseg couples a Gaussian mixture on intensity with Multi-Brain’s diffeomorphic registration. This class of generative model has an inherent accuracy ceiling relative to modern CNN/transformer segmenters on in-distribution tasks with abundant labelled training data. The trade-off is interpretability (each output is a posterior under an explicit model), robustness to scanner/protocol variation (no retraining required when moving between sites), data efficiency (no labels), and a joint segmentation–registration–volumetrics pipeline in a single call; CTseg is therefore best suited to settings where generality and robustness matter more than pushing peak accuracy on a fixed benchmark.

Finally, CTseg takes minutes per scan on a CPU, driven by iterative EM and diffeomorphic registration, versus seconds for GPU-native CNN segmenters. This is acceptable for research throughput and routine offline clinical analysis, but CTseg is not intended for real-time inference.

6.2 Future Work

Valuable extensions include multi-site validation, comparison with deep learning approaches, validation of bone/soft-tissue classes, and evaluation on pathological populations where robust CT segmentation is most needed. Furthermore, CTseg’s generative framework is not inherently specific to brain CT: the tissue probability atlas and spatial normalisation machinery are anatomy agnostic. By training Multi-Brain on, e.g., chest CT scans, the same framework could be applied to different anatomies.

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